

Codefest 6

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

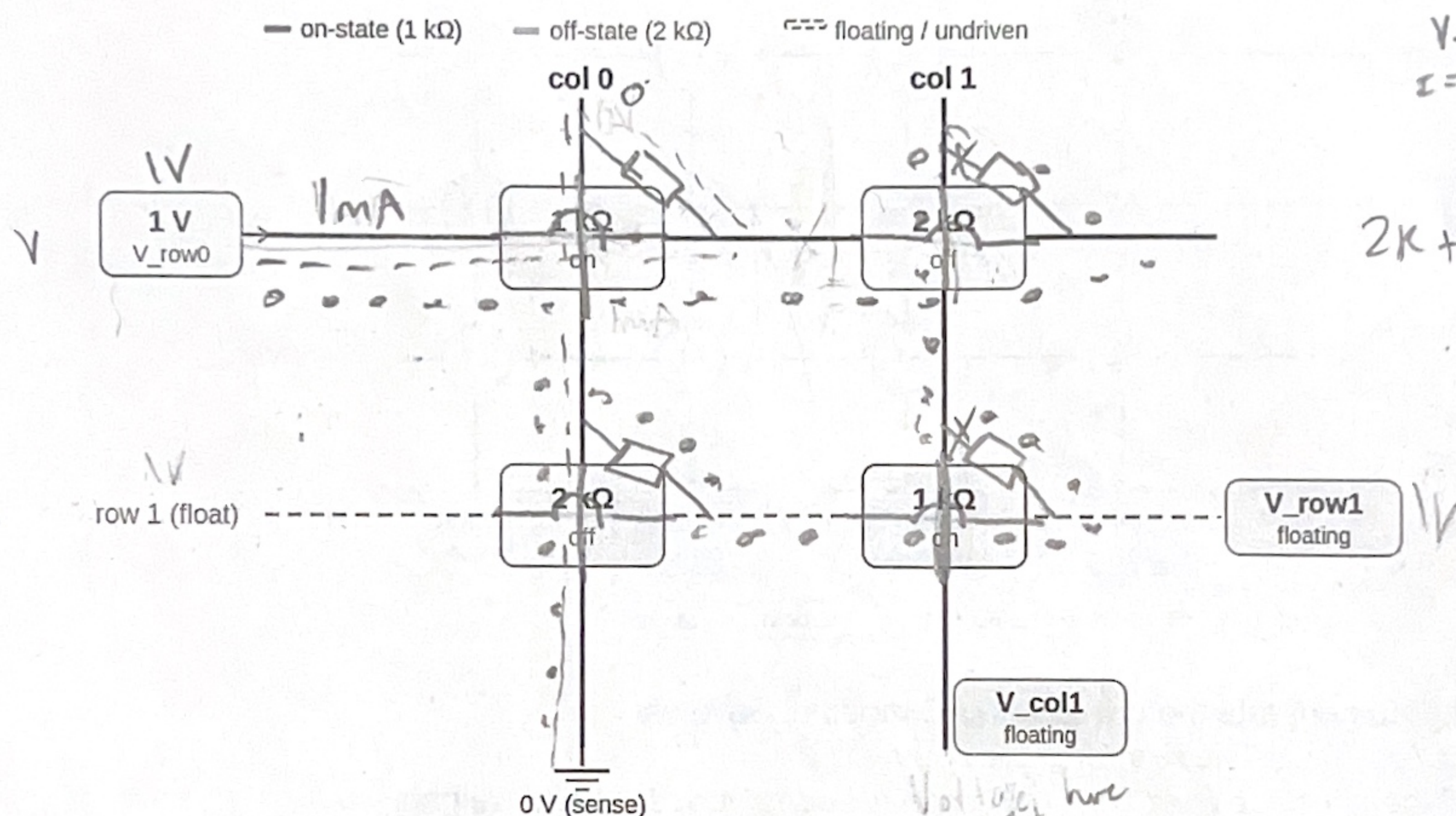
ECE 410/510

Spring 2026

CMAN — Sneak paths in a resistive crossbar

Consider the 2x2 resistive crossbar below. Rows carry input voltages; columns carry output currents. Cell resistances are fixed: $R[0][0] = 1 \text{ k}\Omega$, $R[0][1] = 2 \text{ k}\Omega$, $R[1][0] = 2 \text{ k}\Omega$, $R[1][1] = 1 \text{ k}\Omega$. The low-resistance cells ($1 \text{ k}\Omega$) encode "on" weights; the high-resistance cells ($2 \text{ k}\Omega$) encode "off" weights.

2x2 resistive crossbar — circuit setup



Tasks

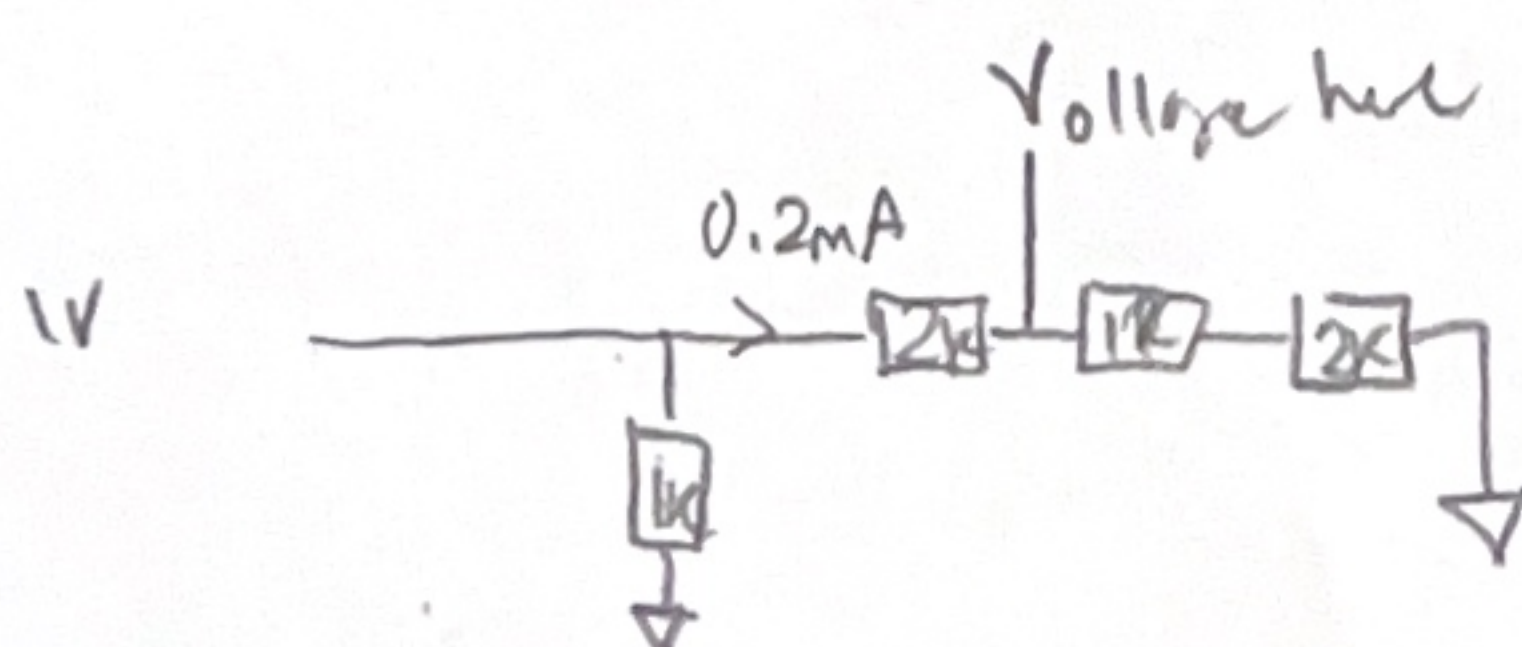
1. Ideal read. Apply $V_{\text{row0}} = 1 \text{ V}$ to row 0. Assume col 0 is held at virtual ground (0 V) for current sensing, and that row 1 and col 1 are also grounded. Compute the ideal output current $I_{\text{col0}} = 1 \text{ mA}$.
2. Sneak path read. Now apply $V_{\text{row0}} = 1 \text{ V}$ to row 0 and hold col 0 at 0 V, but leave row 1 and col 1 floating (undriven). Use Kirchhoff's Current Law (KCL) to find the floating node voltages V_{row1} and V_{col1} , then compute the actual current I_{col0} including the sneak path contribution.
3. In 2–3 sentences, explain why the sneak path current corrupts the intended Matrix Vector Multiplication (MVM) result and what it implies for reading large crossbar arrays.

CMAN NO AI

Deliverable:

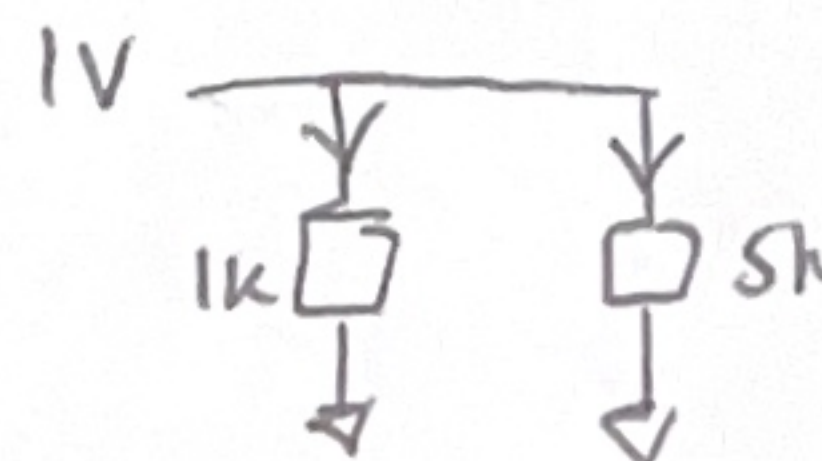
- codefest/cf06/cman_sneak_path.md (or .pdf/.png) containing
 - (a) ideal I_{col0} with calculation;
 - (b) KCL solution for floating node voltages V_{row1} and V_{col1} ;
 - (c) actual I_{col0} with sneak path current itemized;
 - (d) 2–3 sentence explanation of how sneak paths corrupt MVM results.

$$I = \frac{V}{R} = \frac{1\text{V}}{5\text{k}} = 0.2\text{mA}$$



$$V = (0.2\text{mA})(2\text{k}) = 0.4\text{V}$$

$$1 - 0.4\text{V} = 0.6\text{V}$$



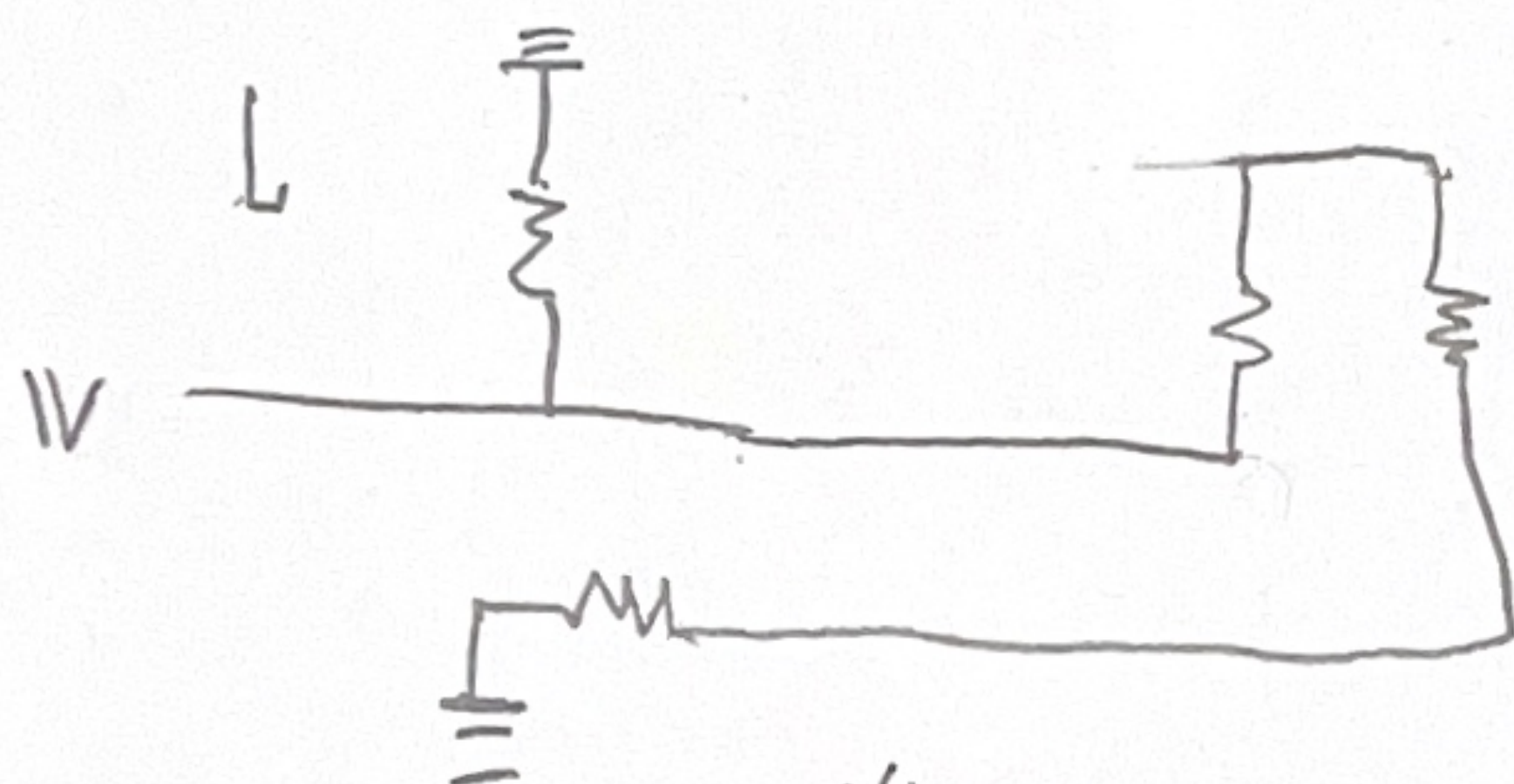
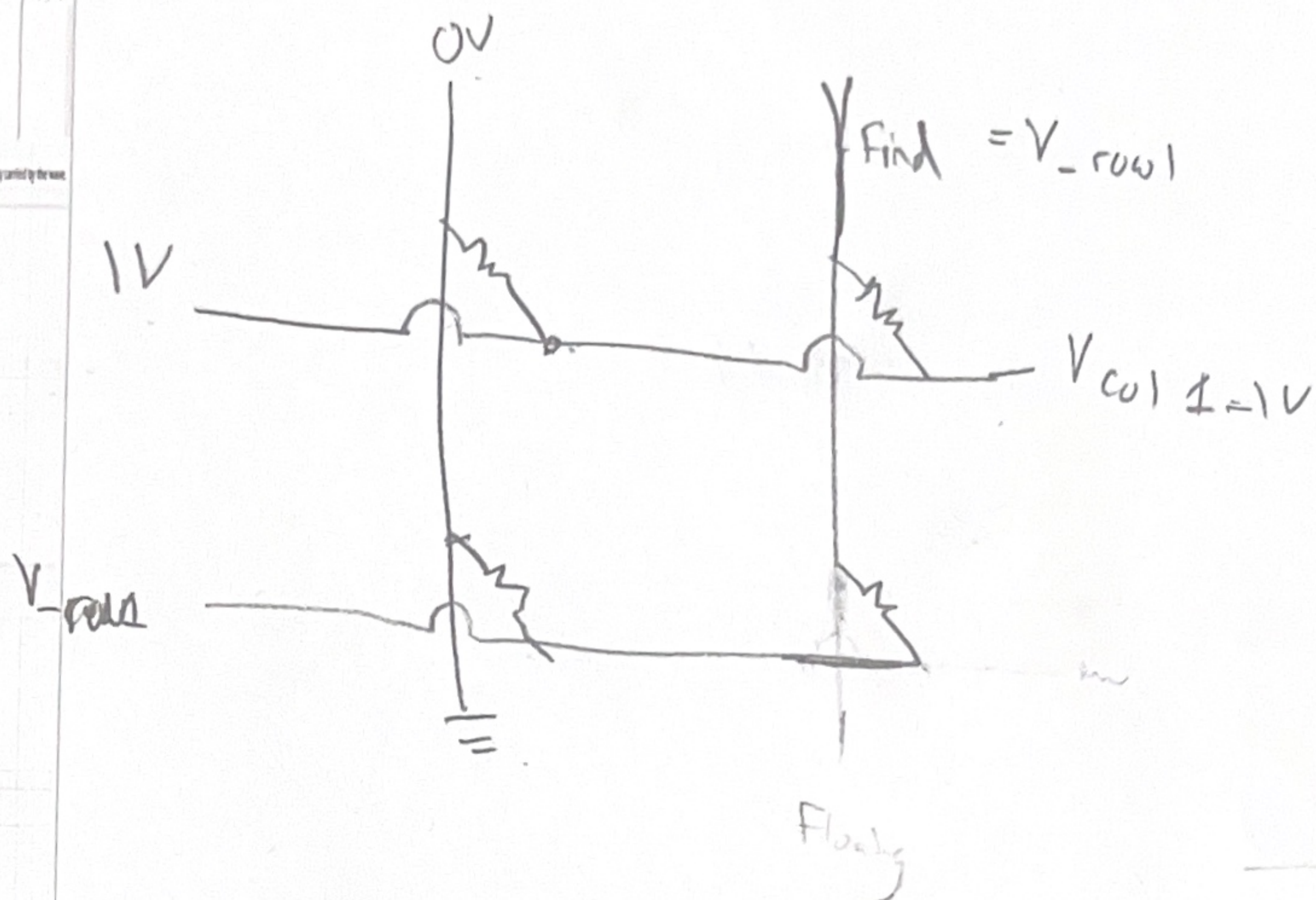
$$\left(\frac{1}{1\text{k}} + \frac{1}{5\text{k}}\right)^{-1} = 833.3\Omega$$

10 points

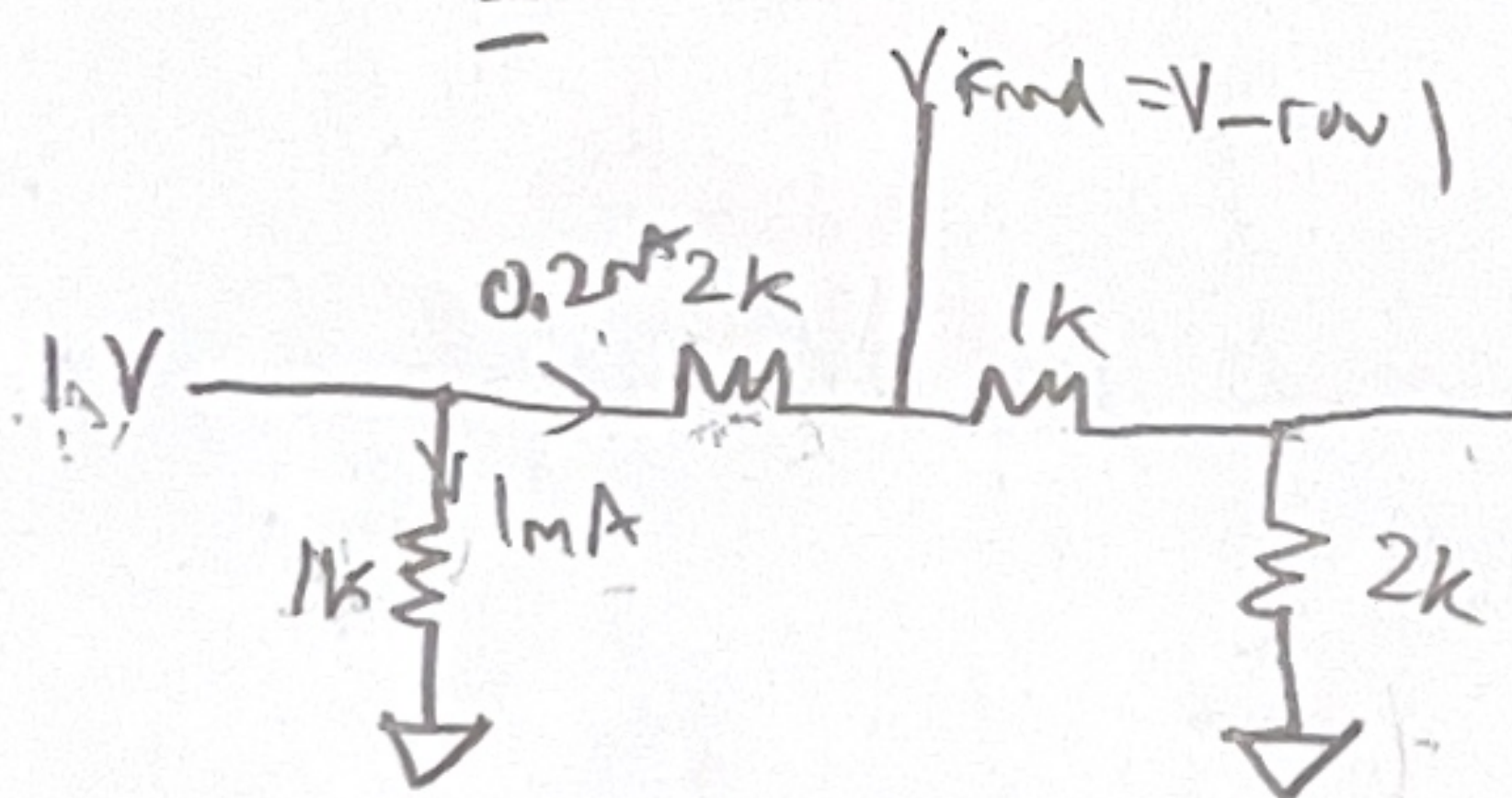
$$E = 2 \cos((x + 10t)) \hat{y} - 2 \sin((x + 10t)) \hat{z} \text{ (V/m)}$$

Determine the direction of wave travel and average power density carried by the wave.

10 points



$$= \frac{1}{\left(\frac{1}{1k} + \frac{1}{5k}\right)}$$



$$V_{row1} = -(0.2m)(2k) + (0.2m)(1k) + 1$$

$$= -0.4 - 0.200 + 1$$

$$= 0.400 \text{ V}$$

$$V = IR$$

$$I = \frac{V}{R} = \frac{1V}{2k + 1k + 2k} = 0.2 \text{ mA}$$

$$V_{drop 2k} = (0.2 \text{ mA})(2k) = 0.4 \text{ V}$$

$$1V - V_{drop 2k} = V_{Find}$$

$$V_{Find} = 1V - 0.4$$

$$= 0.6 \text{ V}$$